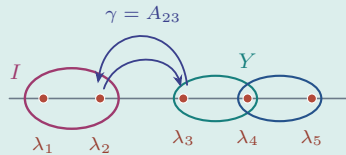


two branch points collide



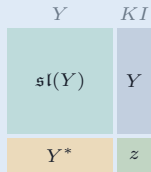
$$E_a = Y \oplus KI \text{ with } Y \cong E_{a'};$$

doubled braids: $E_{a'}$ on Y , ζ^j on I

$$\gamma(I) = I + (1 - t_1)t_2 I_{23}, \text{ so}$$

$$\gamma KI \neq KI \text{ and } \gamma Y \neq Y$$

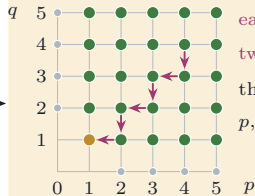
two copies of $SL(Y)$



$S = SL(Y) \times 1$
and $\gamma S \gamma^{-1}$
generate
 $SL(E_a)$

four distinct $\mathfrak{sl}(Y)$ -modules,
as $\dim Y \geq 3$; no common invariant
subspace, so $\mathfrak{t} \supseteq Y, Y^*$, and
 $[Y, Y^*] \ni z$ gives $\mathfrak{t} = \mathfrak{sl}(E_a)$

induction on $n = p + q$



each arrow merges
two branch points;
the order m and
 $p, q \geq 1$ are kept

- $n = 2$: a transvection
- $n \geq 3$: two hyperplanes, by induction
- $p = 0$ or $q = 0$: finite monodromy